ROYAL UNIVERSITY OF PHNOM PENH

Faculty of Engineering

Modernizing Cambodia's Tax Data Warehouse: A Data Vault 2.0 and Medallion Architecture Approach

A Final Report — Master of Science in Data Science and Engineering

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Outline

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Background

GDT Digital Transformation

Cambodia's General Department of Taxation (GDT) is modernizing its digital systems to improve tax collection, compliance monitoring, and revenue forecasting. With thousands of taxpayers filing monthly and annual declarations, transactional data volumes continue to grow rapidly.

Growing Tax Data Volumes

Financial institutions, government agencies, and regulatory bodies increasingly rely on historical tax data for economic policy, compliance enforcement, and resource allocation. This demands an enterprise-grade data warehouse capable of handling large-scale analytical workloads.

Kimball Star Schema Limits

The Kimball dimensional model [3] struggles when source systems change frequently. Star schema requires significant restructuring when new tax types or reporting requirements are introduced, causing costly downtime and redevelopment effort in government IT environments.

Inmon 3NF Limitations

The Inmon corporate information factory [7] produces complex ETL pipelines and slow analytical queries. Its normalized Enterprise Data Warehouse approach lacks the agility needed for modern data integration, parallel development, and evolving regulatory compliance requirements.

Data Vault 2.0 [1] provides the needed agility, auditability, and scalability

Problem Statement

CRITICAL

Schema Rigidity

Star schema models [3] require significant restructuring when source systems change, which happens frequently as government regulations evolve. When a new tax type is introduced or reporting requirements change, the entire dimensional model may need redesign, causing costly downtime.

CRITICAL

Historical Tracking

Traditional approaches struggle to maintain complete historical records. Regulatory bodies require full audit trails showing what data was recorded, when it changed, and previous values. SCD Type 1 overwrites history entirely, while SCD Type 2 [3] requires upfront design decisions.

HIGH

Scalable ETL

Growing tax data volumes demand scalable ETL processes handling both full and incremental loads efficiently. Many implementations rely on manual processes lacking error handling, logging, and recovery mechanisms. Failed loads have no systematic way to identify or restart.

HIGH

Business Rules Gap

There is a gap between raw operational data and business-ready analytical information. Tax administrators need derived metrics such as compliance scores, payment behavior patterns, and revenue trends requiring complex business logic applied consistently across the data warehouse.

These challenges motivate the Data Vault 2.0 + Medallion Architecture approach

Challenges: The Four V

V	Dimension	GDT Challenge	Architectural Response
V1	**Volume**	GDT retains all taxpayer records for legal audit with no deletion. Volume grows continuously and never shrinks.	Watermark-based incremental ETL processes only changed records per cycle. Insert-only Bronze keeps query time constant.
V2	**Velocity**	Monthly deadlines create sharp ingestion spikes with thousands of submissions in days. Daily payments add secondary bursts.	ETL_Watermark delta capture extracts only modified records. 49-step pipeline with Propagate=False error isolation.
V3	**Variety**	Three data categories: lookup, reference, and transaction across 9 tables. Future integration with customs, e-invoicing, banking.	DV2.0 Hub-Satellite-Link [1] separates keys from attributes. New sources integrate via Hubs with zero restructuring.
V4	**Veracity**	Incorrect compliance scores trigger unwarranted audits with legal consequences. Bronze errors propagate silently through Silver and Gold.	Source DB Verification with zero violations across 57,990 records. HashDiff + ETL_StepLog ensure data integrity.

Aim & Objectives

Research Aim: Design & implement a complete DV2.0 [1] + Medallion [2] data warehouse for Cambodia tax administration

O1 — Medallion Architecture

Design a 9-table source database simulating Cambodia's tax system and implement a 4-layer medallion architecture (Staging, Bronze, Silver, Gold) using Data Vault 2.0 Hub-Satellite-Link modeling across 6 databases on SQL Server 2025.

O3 — POC Validation

Validate through 4 proof-of-concept demos: Schema Rigidity (zero-impact satellite addition), Historical Tracking (insert-only audit trail), Scalable ETL (step-level error recovery), and Business Rules (compliance scoring via Business Vault).

O2 — ETL Control Framework

Develop 60 SSIS packages (3 Master, 8 Orchestrator, 49 Child) supporting full and incremental loading with watermark-based change detection. Implement batch-and-step logging, step-level error recovery, and SCD2 history tracking across all layers.

O4 — Gold Layer & Implementation Guide

Create business vault constructs (PIT, Bridge, Business tables) and produce a gold-layer star schema (7 Dims, 4 Facts) for BI dashboards. Provide a Technical Implementation Guide with full SSIS package inventory for reproducibility.

Research Gap

Theoretical Focus

Most Data Vault 2.0 studies remain theoretical or conceptual. Yessad and Labiod [9] present a comparative study of Inmon, Kimball, and Data Vault but provide no complete end-to-end implementation or reproducible proof-of-concept evaluation with measurable outcomes.

Government Tax Domain Gap

No published end-to-end DV2 + Medallion Architecture implementation exists for government tax administration in academic literature. Existing work focuses on telecommunications or healthcare domains where regulatory compliance needs differ significantly.

No End-to-End Framework

Existing studies focus on individual aspects — modeling patterns, ETL performance, or schema evolution — rather than presenting a complete end-to-end implementation spanning all four medallion layers (Staging, Bronze, Silver, Gold) with integrated ETL control frameworks.

Limited ETL Governance

Few studies address integrated ETL control frameworks combining batch logging, step tracking, watermark-based incremental loading, and error recovery within a Data Vault architecture. Production-grade pipeline observability remains largely unaddressed in academic research.

Gap: No published DV2 + Medallion implementation for government tax systems

Contributions

Complete 4-Layer Implementation

A complete, reproducible four-layer medallion architecture [2] (Staging, Bronze, Silver, Gold) using Data Vault 2.0 [1] on SQL Server 2025, comprising 67 database objects across six databases and 60 SSIS packages. This is the first published end-to-end DV2 + Medallion implementation for government tax administration.

Integrated ETL Control Framework

An integrated ETL control framework with batch logging, step tracking, watermark-based incremental loading, and error handling providing production-grade pipeline observability. The framework demonstrates step-level recovery and achieves effective change detection with SAT versioning and SCD2 history tracking.

Implementation Guide & DB Verification

A comprehensive Technical Implementation Guide with complete SSIS package inventory and source database verification. All source code publicly available on GitHub (github.com/mvsotso/dv2-tax-data-warehouse-using-medallion) for full research reproducibility.

POC Evaluation

A proof-of-concept evaluation demonstrating how Data Vault 2.0 addresses four key challenges in Cambodia tax data warehousing: schema flexibility, historical tracking, scalable ETL error recovery, and business rules derivation. Each POC validated with measurable before/after SQL evidence.

First end-to-end DV2 + Medallion implementation for Cambodia tax administration

Literature Review

Year	Study	Key Contribution	Limitation / Gap	Our Approach
Data Vault 2.0 & Medallion Architecture				
2015	Linstedt [1]	Hub/Link/Satellite modeling; hash keys; insert-only audit trail	Foundation methodology; no layered architecture pattern	**DV2 + 4-layer Medallion (Staging→Bronze→Silver→Gold)**
2025	Databricks [2]	Medallion (Bronze/Silver/Gold) for lakehouse pattern [2]	Cloud-native focus; no DV2 integration	**DV2 structures mapped to each Medallion layer**
ETL Frameworks & Data Quality				
2002	Vassiliadis et al.	ETL conceptual modeling; pipeline optimization	No DV2 context; no error recovery	**Control framework + watermark + error isolation**
Related Work — Data Vault Implementations				
2016	Yessad & Labiod [9]	3-way comparison (Inmon/Kimball/DV)	No implementation; purely theoretical	**Complete impl. with POC evidence**
2016	Naamane & Jovanovic [10]	Empirical study with telecom data	No medallion arch.; no Gold layer	**4-layer medallion + Gold star schema**
2024	Vines [11]	TPC-DS benchmark; performance focus	No domain-specific; no ETL control	**Cambodia tax domain + ETL framework**
2025	Singh [12]	Resilience & adaptability analysis	Theoretical analysis; no implementation	**Full POC with measurable evidence**
2026	**This Study**	**DV2 + Medallion + ETL + POC evaluation**	**Simulated data (verified, 12-check)**	**First gov. tax DV2 end-to-end impl.**

Source Database & Sample Data

TaxSystemDB — 9 Source Tables (3 Lookup + 3 Reference + 3 Transaction)

Table	Type	Description	Records
Category	Lookup	Taxpayer categories (Individual, SME, Corp)	**5**
Structure	Lookup	Business structure types (Individual, Partnership, Corp)	**5**
Activity	Lookup	Business activity sectors (Manufacturing, Service, etc.)	**5**
Taxpayer	Reference	Registered taxpayers with TaxID, business name	**1,000**
Officer	Reference	Tax collection officers with department	**10**
Owner	Reference	Business owners with ownership percentages	**1,956**
MonthlyDeclaration	Transaction	Monthly tax declarations with revenue, tax, penalties	**38,491**
Payment	Transaction	Tax payments linked to declarations	**13,733**
AnnualDeclaration	Transaction	Annual tax declarations with yearly summaries	**2,785**

Data Generator (02_TransactionData.sql): 1,000 taxpayers, 4 years (2020–2023), ~80% monthly filing rate, 57,990 records. Verified: 10 FKs, 19 CHECK constraints, 35 indexes, 0 orphans, 0 NULLs (see Source Database Verification document).

Analytical Questions

■ Strategic Analytics

01 Revenue Trends

What are the monthly and annual tax revenue patterns by taxpayer category, business structure, and activity type across 2020–2023?

Gold FACT_MonthlyDeclaration

02 Compliance Scoring

How can taxpayer compliance be measured using declaration timeliness, payment rates, and penalty frequency?

Silver BUS_ComplianceScore

03 Payment Behavior

What are the payment completion rates and outstanding balances by taxpayer segment and declaration period?

Gold FACT_Payment

04 Risk Assessment

Which taxpayers show declining compliance or late declaration trends that require enforcement action?

Gold FACT_DeclarationLifecycle

These questions drive the transformation: Source OLTP → Staging → Bronze (DV2.0) → Silver (Business Vault) → Gold (Star Schema)

System Architecture

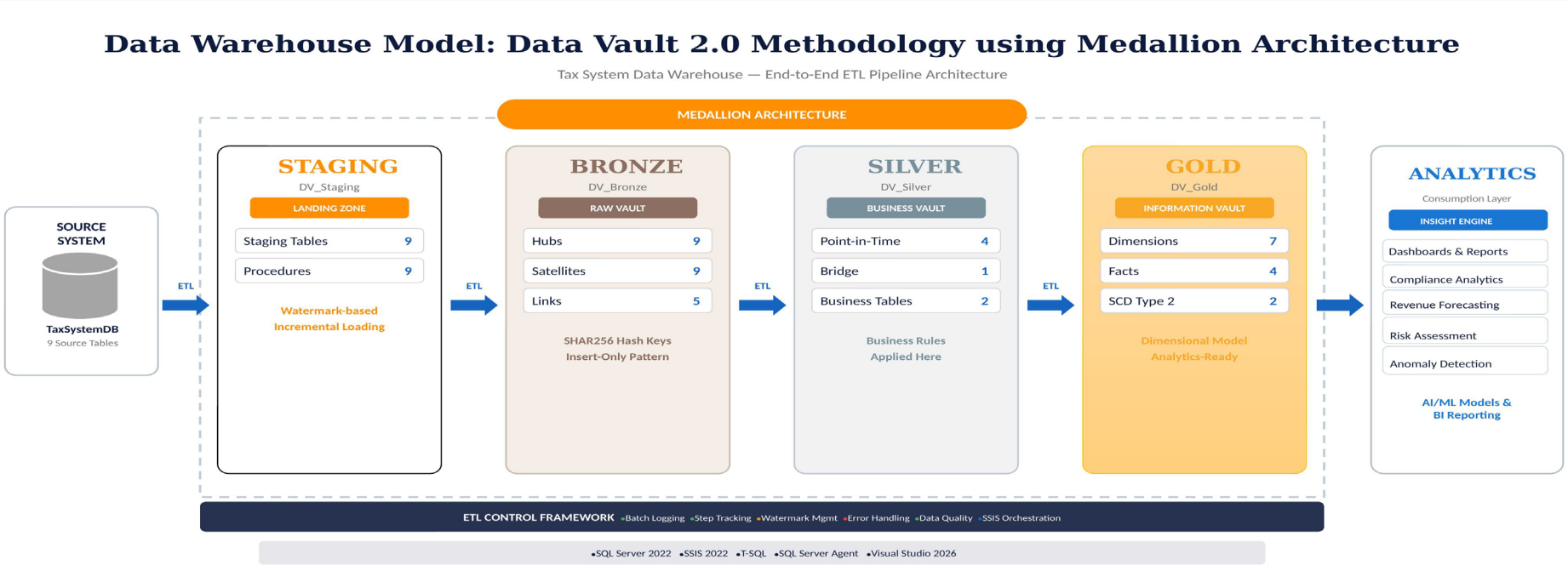

Figure 3.1: Data Warehouse Model: Data Vault 2.0 Methodology using Medallion Architecture

Technology Stack

Software Specification

Technology	Version	Purpose
SQL Server 2025	Developer Edition (RTM 17.0)	Database engine, SSIS runtime, SQL Server Agent
SSIS	Included with SQL Server 2025+	ETL package execution and orchestration [4]
T-SQL	—	SSIS packages, Data Flow tasks, ETL logic
SQL Server Agent	Included with SQL Server 2025	Job scheduling for automated pipeline execution
Visual Studio Community	2026	SSIS package design and development (IDE)

Deployment Environment

Component	Specification	Details
Cloud Platform	Google Cloud Compute Engine	e2-standard-4 instance
CPU	4 vCPUs (2 cores)	Intel Broadwell or newer
Memory	16 GB RAM	12 GB allocated to SQL Server
Storage	128 GB SSD	Persistent disk, single volume
Operating System	Windows Server 2025 Datacenter	Server with Desktop Experience

Implementation — Four Layers

Staging (9)

- Full: TRUNCATE → INSERT
- Incr: Watermark delta
- **1:1 source mapping**

Bronze (23)

- 9 Hubs: SHA-256 hash keys
- 9 SATs: HashDiff + SCD2
- 5 Links: Relationships
- **Insert-only audit trail [1]**

Silver (6)

- 3 PIT: Temporal snapshots
- 1 Bridge: Pre-joined
- **2 BUS: Compliance scores**

Gold (11)

- 7 DIMs: SCD1 + SCD2
- 4 FACTs: Monthly, Payment
- Snapshot, Audit
- **Star schema for BI**

Master Orchestration — 49 Steps

Init Batch → Staging (9) → Hubs (9) → SATs (9) → Links (5) → Silver (6) → Gold (11) → Close Batch

Analytical Questions → Implementation

How each analytical question is answered by the DV2 + Medallion architecture (Gold star schema from DV_Gold database)

01 Revenue Trends
Monthly/annual tax revenue patterns by taxpayer category, structure, and activity (2020–2023)
→ **Gold FACT_MonthlyDeclaration**
Star schema fact joins DIM_Taxpayer, DIM_Category, DIM_Structure, DIM_Activity for BI reporting

02 Compliance Scoring
Measure compliance via declaration timeliness, payment rates, and penalty frequency
→ **Silver BUS_ComplianceScore**
Business Vault derives compliance ratings from declaration timeliness, payment behavior, penalty history

03 Payment Behavior
Payment completion rates and outstanding balances by taxpayer segment and period
→ **Gold FACT_Payment**
Fact table with payment amounts, method, status joined to DIM_Taxpayer and DIM_PaymentMethod

04 Risk Assessment
Identify taxpayers with declining compliance or late declaration trends for enforcement
→ **Gold FACT_DeclarationLifecycle**
Tracks declaration submission, status changes, late filing patterns with DIM_Taxpayer and DIM_Status

Data Flow Through Layers

Source OLTP
TaxSystemDB
9 tables | 58K records

→

Staging
DV_Staging
9 tables

→

Bronze (DV2.0)
DV_Bronze
9 Hubs | 9 SATs | 5 Links

→

Silver (BV)
DV_Silver
3 PIT | 1 Bridge | 2 BUS

→

Gold (Star)
DV_Gold
7 DIM | 4 FACT

67 database objects across 6 databases • 60 SSIS packages • Full/Incremental load with watermark-based change detection

POC Evidence — 4 Challenges Addressed

C1: Schema Rigidity	Problem: ALTER TABLE impacts multiple dependent objects. DV2 Result: 0 of 67 objects modified, 0 of 60 SSIS packages changed	DV2: CREATE new Satellite only — zero impact on 60 SSIS packages
C2: Historical Tracking	Problem: Limited audit trail, no full version history. DV2 Result: All 3 versions preserved with timestamps (200K → 350K → 505K)	DV2: Insert-only SATs [1] preserve ALL versions with HashDiff + timestamps
C3: Scalable ETL	Problem: Monolithic pipelines, no step-level logging or error recovery. DV2 Result: 7 of 8 steps recovered, 45,065 rows processed despite 1 failure	DV2: ETL control with BatchLog/StepLog/ErrorLog, OnError Event Handlers
C4: Business Rules Gap	Problem: No separation of raw data from derived business logic. DV2 Result: Compliance score 72.5%, risk level MEDIUM for TAX000001	DV2: Silver applies business rules (ComplianceScore); Bronze preserves raw

Figure 5.1 — POC: Schema Flexibility via Hub-Satellite Separation

Figure 5.1 — POC: Schema Flexibility via Hub-Satellite Separation

Scenario: A new attribute "TaxpayerEmail" is required — demonstrating how Data Vault 2.0 accommodates source changes without modifying existing objects

CURRENT BRONZE LAYER STRUCTURE (BEFORE CHANGE)

Object	Type	Key Columns	Purpose
HUB_Taxpayer	Hub	HUB_Taxpayer_HK, TaxID	Business key — never changes
SAT_Taxpayer	Satellite	HUB_Taxpayer_HK, SAT_LoadDate	Descriptive: Name, Revenue, Category
LNK_TaxpayerDeclaration	Link	HUB_Taxpayer_HK, HUB_Declaration_HK	Relationship — never changes
LNK_TaxpayerOfficer	Link	HUB_Taxpayer_HK, HUB_Officer_HK	Relationship — never changes

ADDING NEW ATTRIBUTE: CREATE NEW SATELLITE (ONLY ACTION REQUIRED)

```
-- New Satellite — no existing objects modified
CREATE TABLE DV_Bronze.dbo.SAT_Taxpayer_Contact (
    HUB_Taxpayer_HK    VARBINARY(32)  NOT NULL,
    SAT_LoadDate       DATETIME       NOT NULL,
    SAT_EndDate        DATETIME       NULL,
    SAT_RecordSource   VARCHAR(50)    NOT NULL,
    SAT_HashDiff       VARBINARY(32)  NOT NULL,
    TaxpayerEmail      VARCHAR(200),
    TaxpayerPhone      VARCHAR(50),
    CONSTRAINT PK_SAT_Taxpayer_Contact PRIMARY KEY (HUB_Taxpayer_HK, SAT_LoadDate),
    CONSTRAINT FK_SAT_TpContact_HUB FOREIGN KEY (HUB_Taxpayer_HK)
        REFERENCES DV_Bronze.dbo.HUB_Taxpayer (HUB_Taxpayer_HK)
);
```

AFTER CHANGE — IMPACT ASSESSMENT

Object	Type	Impact
HUB_Taxpayer	Hub	✅ No change — business key unchanged
SAT_Taxpayer	Satellite	✅ No change — existing attributes intact
LNK_TaxpayerDeclaration	Link	✅ No change — relationship unchanged
LNK_TaxpayerOfficer	Link	✅ No change — relationship unchanged
SAT_MonthlyDecl	Satellite	✅ No change — independent satellite
All SSIS Packages (60)	ETL	✅ No change — existing packages untouched
SAT_Taxpayer_Contact	**New Satellite**	✅ **Created independently, linked via HUB_Taxpayer_HK**

Result: New attribute added by creating 1 new Satellite table + 1 new SSIS package. Zero existing objects modified. The Hub-Satellite separation pattern [1][5] ensures that descriptive attribute changes never impact business keys or relationships.

Figure 5.2 — POC: Full Historical Tracking via Insert-Only Satellites

Figure 5.2 — POC: Full Historical Tracking via Insert-Only Satellite Pattern

Scenario: Taxpayer TAX000001's EstimatedAnnualRevenue changes 3 times — demonstrating complete audit trail preservation

HASHDIFF-BASED CHANGE DETECTION (BRONZE LAYER STORED PROCEDURE)

```sql
-- From 06_ETL_Bronze_Procedures.sql: usp_Load_SAT_Taxpayer
-- Step 1: End-date existing current records where HashDiff changed
UPDATE sat SET sat.SAT_EndDate = GETDATE()
FROM DV_Bronze.dbo.SAT_Taxpayer sat
INNER JOIN DV_Staging.dbo.STG_Taxpayer stg
    ON sat.HUB_Taxpayer_HK = stg.HUB_Taxpayer_HK
WHERE sat.SAT_EndDate IS NULL
  AND sat.SAT_HashDiff <> stg.SAT_HashDiff;   -- Only when attributes actually changed

-- Step 2: Insert new version for changed records
INSERT INTO DV_Bronze.dbo.SAT_Taxpayer (HUB_Taxpayer_HK, SAT_LoadDate, SAT_HashDiff, ...)
SELECT stg.HUB_Taxpayer_HK, GETDATE(), stg.SAT_HashDiff, ...
FROM DV_Staging.dbo.STG_Taxpayer stg
LEFT JOIN DV_Bronze.dbo.SAT_Taxpayer sat
    ON stg.HUB_Taxpayer_HK = sat.HUB_Taxpayer_HK AND sat.SAT_EndDate IS NULL
WHERE sat.HUB_Taxpayer_HK IS NULL              -- New record (no current version)
   OR sat.SAT_HashDiff <> stg.SAT_HashDiff;   -- Changed record (different hash)
```

QUERY RESULT — SAT_TAXPAYER: ALL 3 VERSIONS PRESERVED FOR TAX000001

HUB_Taxpayer_HK (first 8)	LegalBusinessName	Revenue	SAT_LoadDate	SAT_EndDate	Version
0x3A7F29B1...	Business 1	200,000	2026-02-22 08:00:15	2026-02-22 08:05:30	V1 (Historic)
0x3A7F29B1...	Business 1	350,000	2026-02-22 08:05:30	2026-02-22 08:10:45	V2 (Historic)
0x3A7F29B1...	Business 1 (AMENDED)	**505,000**	2026-02-22 08:10:45	*NULL*	**V3 (Current)**

Result: All 3 versions preserved with precise timestamps. SAT_EndDate = NULL identifies the current version. The insert-only pattern [1][5] with HASHBYTES('SHA2_256') change detection ensures only genuine changes are recorded — unchanged records are not duplicated. Government auditors can query any point-in-time using SAT_LoadDate.

Figure 5.3 — POC: ETL Control Framework with Error Recovery

Figure 5.3 — POC: ETL Control Framework with Step-Level Logging and Error Recovery

Scenario: Load_STG_Payment fails with timeout during Staging layer execution — demonstrating error isolation and recovery capability

ETL_BATCHLOG — PIPELINE-LEVEL STATUS

BatchID	BatchType	BatchStatus	BatchStartTime	BatchEndTime	Duration
42	Full Load	⚠ **Partial**	2026-02-22 08:00:00	2026-02-22 08:02:01	121 sec

ETL_STEPLOG — STEP-LEVEL DETAIL (INDEPENDENT ERROR HANDLING PER PACKAGE)

StepLogID	StepName	StepStatus	RecordsProcessed	Duration (sec)	ErrorMessage
201	Load_STG_Category	✓ Success	5	1.2	
202	Load_STG_Structure	✓ Success	5	0.8	
203	Load_STG_Taxpayer	✓ Success	1,000	4.5	
204	**Load_STG_Payment**	**✗ Failed**	0	60.0	**Execution Timeout Expired [Source: TaxSystemDB.dbo.Payment]**
205	Load_STG_MonthlyDecl	✓ Success	38,487	18.3	
206	Load_STG_AnnualDecl	✓ Success	3,605	5.1	
207	Load_STG_Officer	✓ Success	10	0.3	
208	Load_STG_Owner	✓ Success	1,953	3.2	

ETL_ERRORLOG — DETAILED ERROR CAPTURE (ONERROR EVENT HANDLER)

ErrorLogID	BatchID	StepName	ErrorMessage	ErrorTime
15	42	Load_STG_Payment	Execution Timeout Expired. The timeout period elapsed prior to completion of the operation. [Source: TaxSystemDB.dbo.Payment, OLE DB provider]	2026-02-22 08:01:05

Result: The SSIS OnError Event Handler [4] captured the error via System::ErrorDescription and logged it to ETL_ErrorLog. Key outcomes: (1) Failed step identified instantly: Load_STG_Payment. (2) 7 of 8 steps completed successfully — independent error handling per package ensures failures don't cascade. (3) 45,065 rows processed across successful steps. (4) **Recovery: re-run only Load_STG_Payment** — no need to restart the entire pipeline.

Figure 5.4 — POC: Business Rules Applied via Silver + Gold Layers

Figure 5.4 — POC: Business Rules Applied via Silver (Business Vault) + Gold (Star Schema)

Scenario: Derived analytics for TaxID TAX000001 — demonstrating pre-computed compliance metrics from raw Bronze data

SILVER LAYER: BUS_COMPLIANCESCORE (COMPUTED FROM BRONZE HUB + SATELLITE + LINK)

```sql
-- From 07_ETL_Silver_Procedures.sql: usp_Load_BUS_ComplianceScore
-- ComplianceScore = 50% FilingOnTimeRate + 50% PaymentOnTimeRate
SELECT hub.HUB_Taxpayer_HK,
       ISNULL(filing.OnTimeRate, 0) * 0.50 + ISNULL(payment.OnTimeRate, 0) * 0.50 AS ComplianceScore,
       filing.OnTimeRate  AS FilingOnTimeRate,
       payment.OnTimeRate AS PaymentOnTimeRate,
       penalties.PenaltyCount
FROM DV_Bronze.dbo.HUB_Taxpayer hub
OUTER APPLY (/* filing on-time calculation */) filing
OUTER APPLY (/* payment completion rate */) payment
OUTER APPLY (/* penalty count */) penalties;
```

72.5%	78.3%	66.7%	MEDIUM	2
Compliance Score	Filing On-Time Rate	Payment On-Time Rate	Risk Classification	Penalty Count

GOLD LAYER: FACT_MONTHLYDECLARATION (STAR SCHEMA — BI DASHBOARD READY)

DIM_Taxpayer	DIM_Status	Month	GrossRevenue	TaxAmount	TotalAmount	PaymentAmount	Coverage %
TAX000001	Approved	Jan 2023	45,000	6,750	6,750	6,750	100%
TAX000001	Approved	Feb 2023	52,000	7,800	7,800	7,800	100%
TAX000001	Rejected	Mar 2023	38,000	5,700	5,700	0	0%
TAX000001	Approved	Apr 2023	61,000	9,150	9,150	9,150	100%
TAX000001	Pending	May 2023	55,000	8,250	8,250	0	0%
TAX000001	Approved	Jun 2023	48,000	7,200	7,200	7,200	100%

Result: The Silver layer (Business Vault) computes derived metrics using consistent business rules [1][2] — compliance score formula: 50% filing on-time rate + 50% payment on-time rate. The Gold layer (Star Schema) organizes these into dimensional models with DIM/FACT joins for BI dashboard consumption. Tax administrators get instant, consistent analytics without writing ad-hoc queries across raw Bronze tables.

Conclusion

This research successfully designed, implemented, and evaluated a complete Data Vault 2.0 data warehouse for the Cambodia tax system using medallion architecture on Microsoft SQL Server.

- Complete 4-layer architecture: 67 objects, 60 SSIS packages, 49 ETL steps across 6 databases
- ETL pipeline: Full and incremental loading with watermark-based change detection and per-step error isolation
- Gold layer star schema provides a dimensional model for BI reporting
- Performance benchmarking and scalability analysis identified as future work
- POC validation: All 4 challenges (schema rigidity, history, scalable ETL, business rules) addressed
- All source code, SQL scripts, SSIS packages, and documents publicly available on GitHub for full reproducibility

  `github.com/mvsotso/dv2-tax-data-warehouse-using-medallion`

Future Work & Recommendations

1. Real-time processing with CDC (Change Data Capture) for critical compliance monitoring tables

2. Production deployment with real GDT data — data quality validation, security classification, access controls

3. Performance benchmarking: ETL load times, query performance (Gold vs Raw Vault), scalability analysis with columnstore indexes

4. BI dashboard integration (Power BI, Metabase) consuming Gold layer star schema for tax analytics

5. Multi-source integration: customs, banking, e-invoicing via Same-As Links & Multi-Active Satellites

6. Advanced analytics: ML models on Gold layer for compliance risk scoring, revenue forecasting, fraud detection, and anomaly detection

References

Core Methodology

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Full reference details available in the Final Report, Chapter: References

Thank You

Questions & Discussion

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